CHANNA KAVISHKA SADARUWAN

SOFTWARE ENGINEERING INTERN • TRAINEE SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Final-year BSc (Hons) Software Engineering student with practical experience engineering full-stack web applications, interactive 3D WebGL interfaces, enterprise Java backend services, and explainable AI systems. Proven track record of shipping **3 live production deployments** across Vercel, custom domains, and cloud environments. Proficient in **Java, Spring Boot, JavaScript (ES6+), React.js, Node.js, Express, Python, MySQL, and MongoDB**, with strong foundational knowledge in relational database design, RESTful architectures, and clean code principles. Seeking a Software Engineering Internship / Trainee role to contribute immediately to production feature delivery while expanding enterprise architectural capabilities.

TECHNICAL SKILLS

Languages:	Java (Core & OOP), JavaScript (ES6+), Python, SQL (MySQL)
Frontend & 3D:	React.js, Three.js (WebGL), GSAP ScrollTrigger, Lenis Smooth Scroll, WebXR (AR), Tailwind CSS, HTML5/CSS3
Backend & APIs:	Node.js, Express.js, Java Spring Boot, RESTful APIs, Spring Data JPA, Hibernate, JWT Authentication, MVC
Databases & AI Tools:	MySQL, MongoDB, Scikit-learn, XGBoost, LightGBM, SHAP, Git, GitHub, Stripe, Vercel
Core Competencies:	Full-Stack Web Engineering, Relational Database Modeling, System Architecture, Agile/Scrum, Problem Solving

TECHNICAL PROJECTS

Café Nuwara — Interactive 3D Web Application

Live Demo ↗ • GitHub ↗

Tech Stack: JavaScript (ES6+), Three.js (WebGL), GSAP ScrollTrigger, Lenis, Tailwind CSS, Vercel

- **Architected a real-time 3D web application** utilizing Three.js and WebGL rendering pipeline to deliver interactive model manipulation with steady 60 FPS viewport performance.
- **Engineered scroll-synchronized animations** with GSAP ScrollTrigger and Lenis smooth scrolling, mapping camera transformations and layer-by-layer asset anatomy breakdowns to user scroll depth.

iFurnish Shop — Full-Stack 3D E-Commerce & WebXR Spatial Preview

Live Demo ↗ • GitHub ↗

Tech Stack: MERN Stack (MongoDB, Express.js, React.js, Node.js), Three.js, WebXR (AR), Stripe, Vercel

- **Developed a full-stack furniture commerce application** featuring real-time 3D WebGL material customization and WebXR Augmented Reality for in-room spatial projection on mobile.
- **Built secure Node/Express REST APIs** with JWT authentication, Stripe payment processing, persistent cart management, and dynamic MongoDB filtering (BSc Final Thesis).

AnimeVerse — Dynamic Media Streaming & Discovery Platform

Live Demo ↗ • GitHub ↗

Tech Stack: JavaScript (ES6+), Node.js, Express.js, HLS.js, ArtPlayer, Tailwind CSS, Custom Domain Hosting

- **Engineered a streaming platform with an Express.js stream proxy** and client-side caching to resolve video streams, bypass CORS limits, and eliminate redundant network roundtrips.
- **Integrated ArtPlayer and HLS.js for adaptive playback**, implementing debounced real-time query search, responsive dark-mode styling, and production deployment on custom domain (animeverse.lol).

Oceanview Reservation System — Enterprise Hotel Booking Platform

GitHub ↗

Tech Stack: Java, Spring Boot, MySQL, Hibernate, Spring Data JPA, MVC Architecture

- **Developed an enterprise reservation system** using Spring Boot following clean layered MVC, service-layer business logic, and repository abstraction patterns.
- **Engineered relational database models with Hibernate/JPA**, enforcing ACID transactional consistency across room availability checks, reservation booking lifecycles, and check-in/out workflows.

HealthShield AI — Explainable Machine Learning Diagnostic Platform

GitHub ↗

Tech Stack: Python, Streamlit, Scikit-learn, XGBoost, LightGBM, SHAP, Pandas

- **Engineered an explainable machine learning platform in Python & Streamlit** predicting multi-factorial health risks using a soft-voting ensemble of XGBoost, LightGBM, and CatBoost.
- **Integrated SHAP (SHapley Additive exPlanations)** for model interpretability, feature importance analysis, and transparent clinician risk assessment with 5-Fold Cross-Validation.

EDUCATION

BSc (Hons) in Software Engineering — 2023 – 2026 (Expected)
Cardiff Metropolitan University • Undergraduate Degree
Key Coursework: Data Structures & Algorithms, Object-Oriented Software Engineering, Database Systems, Web Application Architectures, Distributed Computing, Software Quality Assurance.

Higher National Diploma (HND) in Software Engineering — Completed
ICBT Campus

Diploma in English — Completed
British Way English Academy • Professional English Communication